

Computer Science (Degree)

CTSE 2025 - MCQ Paper - 100 Questions with Answers and Explanations

- Booklet 2537_25 (A), Series A - 100 objective questions
- Topic spread: Q1-30 programming, data structures and algorithm analysis; Q31-50 digital logic, computer architecture and system software; Q51-67 operating systems and software engineering; Q68-76 databases; Q77-89 networks and web technology; Q90-100 artificial intelligence and machine learning

How to use this document

- Questions are grouped in blocks of ten. Each block of ten questions is followed by a spoiler bar and a page break, then the answers for those ten questions.
- This grouping is deliberate: it lets you attempt ten at a time and check them, instead of turning a page after every single question. Tell me if you would rather have one page break per question.
- Each answer gives the correct option and a short explanation of why it is right, and where useful why the tempting wrong option is wrong.
- Four questions in this paper are genuinely ambiguous as printed. Those answers carry a highlighted note explaining the ambiguity and what the official key most likely says.

Questions 1 to 10

1. Which component mitigates the Von Neumann bottleneck in modern CPUs?
 - A) Larger ALUs
 - B) Cache memory hierarchy
 - C) Increased clock speed
 - D) Wider data buses
2. What is the time complexity of linear search in the worst case?
 - A) $O(1)$
 - B) $O(\log n)$
 - C) $O(n)$
 - D) $O(n^2)$
3. Which register holds the address of the next instruction?
 - A) Accumulator
 - B) Program Counter
 - C) Instruction Register
 - D) Memory Address Register
4. Which data type among the following in C programming language, has the smallest storage size?
 - A) int
 - B) char
 - C) float
 - D) double
5. What is the output of the instruction: `printf("%d", 5 / 2)`, in C?
 - A) 2
 - B) 2.5
 - C) Division Error
 - D) Syntax Error
6. What does the volatile keyword in C signify?
 - A) The variable cannot be optimized by the compiler.
 - B) The variable is constant.
 - C) The variable is stored in ROM.
 - D) The variable is thread-safe.
7. Which header file is required for `malloc()` in C?
 - A) `<stdio.h>`
 - B) `<stdlib.h>`

- C) <memory.h>
- D) <conio.h>

8. What does the C/C++ built-in function strcmp() return if two strings are identical?

- A) 0
- B) 1
- C) -1
- D) Address of the first string

9. What is the scope of a static variable inside a function?

- A) Limited to the function block
- B) Global across the program
- C) Limited to the file where it is defined
- D) Retains value between function calls

10. Which C++ feature ensures type safety for bitwise operations on enums?

- A) constexpr
- B) enum class
- C) volatile
- D) mutable

▼ ANSWERS TO QUESTIONS 1-10 BELOW ▼

Finish all ten before turning the page.

Answers - Questions 1 to 10

1. Answer: B) Cache memory hierarchy

The Von Neumann bottleneck is the limited throughput of the single path between CPU and memory. The cache hierarchy attacks it directly by keeping frequently used instructions and data in fast memory next to the CPU, so most references never traverse that path. Bigger ALUs and higher clock speeds make the CPU faster and therefore make the bottleneck worse; a wider bus helps somewhat but does not reduce the number of accesses.

2. Answer: C) $O(n)$

Linear search examines elements one at a time. In the worst case the target is the last element or absent altogether, so all n elements are compared, giving $O(n)$. $O(1)$ would be the best case, when the target is first.

3. Answer: B) Program Counter

The Program Counter, also called the instruction pointer, holds the address of the next instruction to be fetched, and is incremented automatically during the fetch cycle. The Instruction Register holds the instruction currently being executed, the MAR holds the address currently on the bus, and the accumulator holds ALU operands and results.

4. Answer: B) char

char is defined to be exactly one byte, and `sizeof(char)` is 1 by definition in C. Typical sizes on a modern platform are char 1, int 4, float 4, double 8 bytes.

5. Answer: A) 2

Both operands are integer literals, so C performs integer division and truncates towards zero, giving 2. To obtain 2.5 at least one operand must be floating point, as in `5.0 / 2`, and the format specifier would then have to be `%f`.

6. Answer: A) The variable cannot be optimized by the compiler.

`volatile` tells the compiler that the variable may change outside the program's own control - through a memory-mapped hardware register, an interrupt handler, or another thread - so the compiler must not cache it in a register or optimise away apparently redundant reads. Note that it does not make access atomic, so it is not a substitute for proper synchronisation, which rules out option D.

7. Answer: B) `<stdlib.h>`

`malloc`, `calloc`, `realloc` and `free` are all declared in `<stdlib.h>`, along with `exit`, `atoi` and the random-number functions. `<stdio.h>` provides the input and output functions.

8. Answer: A) 0

`strcmp` returns 0 for equality, a negative value if the first string sorts before the second, and a positive value if it sorts after. The common mistake is to assume it returns true for a match, so the idiom is if `(strcmp(a, b) == 0)`, not if `(strcmp(a, b))`.

9. Answer: A) Limited to the function block

Strictly, scope is where a name is visible, and a static local variable is visible only inside its own function, so A is the technically correct answer to the question as asked. Option D describes its lifetime, or storage duration, which is the whole run of the program - a different property.

Ambiguous as printed - see the explanation above for what the official key most likely records. Worth flagging to your tutor.

10. Answer: B) enum class

enum class, the scoped enumeration introduced in C++11, does not implicitly convert to int, so its enumerators cannot be silently mixed with integers or with other enumerations. That is the type safety in question; using it for bit flags requires explicitly defined operators or a cast.

Questions 11 to 20

11. What is the output of the following code? `int arr[5] = {1, 2}; printf("%d", arr[2]);`

- A) 1
- B) 2
- C) 0
- D) Garbage Value

12. What is the time complexity to insert a node at the head of a singly linked list?

- A) $O(1)$
- B) $O(n)$
- C) $O(\log n)$
- D) $O(n^2)$

13. What is the value of the following expression in C/C++, assuming standard operator precedence? `10 + 5 * 2 / 4 - 1`

- A) 6
- B) 11
- C) 7.5
- D) 10

14. What is the primary purpose of a function prototype (or declaration) in C/C++?

- A) To define the memory location where the function's code resides.
- B) To specify the function's return type, name, and parameter types before its actual definition or use.
- C) To execute the function's code block.
- D) To declare local variables used within the function.

15. In the following expression, what does the `*ptr` evaluate to? `int value = 10; int *ptr = &value;`

- A) The memory address of value
- B) The value 10
- C) The memory address stored in ptr
- D) The size of an integer

16. What potential problem occurs by dereferencing a pointer that has not been initialized or has been assigned NULL?

- A) The program compiles with a warning.
- B) The pointer automatically points to a default memory location.
- C) It results in undefined behavior, often leading to a program crash.
- D) It returns the value 0.

17. What is the typical time complexity for `Push()` and `Pop()` operations on a stack implemented using either an array or a linked list?

- A) $O(n)$
- B) $O(\log n)$
- C) $O(1)$
- D) $O(n^2)$

18. Which of the following applications is best suited for a queue data structure?

- A) Evaluating arithmetic expressions.
- B) Implementing an "Undo" feature in a text editor.
- C) Managing tasks scheduled for execution in order of arrival.
- D) Tracking nested function calls.

19. In tree data structures, a node with no children is called as:

- A) Root Node
- B) Parent Node
- C) Internal Node
- D) Leaf Node

20. Which property must be satisfied by a Binary Search Tree (BST) for every node?

- A) All nodes in the left subtree have keys greater than the node's key, and all nodes in the right subtree have keys less than the node's key.
- B) All nodes in the left subtree have keys less than or equal to the node's key, and all nodes in the right subtree have keys greater than or equal to the node's key.
- C) The height of the left subtree and the right subtree differ by at most 1.
- D) Each node must have exactly two children.

▼ ANSWERS TO QUESTIONS 11-20 BELOW ▼

Finish all ten before turning the page.

Answers - Questions 11 to 20

11. Answer: C) 0

When an array is partially initialised, the remaining elements are implicitly initialised to zero. So arr is {1, 2, 0, 0, 0} and arr[2] is 0. A garbage value would result only from an uninitialised array with no initialiser list at all.

12. Answer: A) $O(1)$

Insertion at the head requires setting the new node's next pointer to the current head and then updating the head pointer - two operations, independent of list length, so $O(1)$. Insertion at the tail of a singly linked list without a tail pointer would be $O(n)$.

13. Answer: B) 11

Multiplication and division bind tighter than addition and subtraction and associate left to right. So $5 * 2 = 10$, then $10 / 4 = 2$ by integer division, then $10 + 2 - 1 = 11$. The trap is $10 / 4$: with integers the fractional part is discarded, which is why the answer is not 7.5.

14. Answer: B) To specify the function's return type, name, and parameter types before its actual definition or use.

A prototype declares the function's signature so the compiler can check the number and types of arguments at every call site, and can apply the correct conversions, even if the definition appears later in the file or in another translation unit.

15. Answer: B) The value 10

ptr holds the address of value; the dereference operator * follows that address and yields the object stored there, which is 10. The address itself is what ptr evaluates to without the asterisk.

16. Answer: C) It results in undefined behavior, often leading to a program crash.

Dereferencing a null or wild pointer is undefined behaviour in both C and C++. In practice it usually causes a segmentation fault, but the standard permits anything, and the program may equally appear to work while corrupting memory - which is what makes it dangerous. It is a run-time fault, not a compile-time warning.

17. Answer: C) $O(1)$

Both operations act only at the top: in an array implementation they index and adjust a top counter; in a linked list they add or remove the head node. Neither involves traversal or shifting, so both are constant time in either implementation.

18. Answer: C) Managing tasks scheduled for execution in order of arrival.

A queue is FIFO, so it naturally serves items in order of arrival - CPU scheduling, print spooling, request buffering. The other three are all LIFO problems and call for a stack: expression evaluation, undo history, and the call stack.

19. Answer: D) Leaf Node

A leaf, also called an external or terminal node, has no children. The root has no parent, and an internal node has at least one child.

20. Answer: B) All nodes in the left subtree have keys less than or equal to the node's key, and all nodes in the right subtree have keys greater than or equal to the node's key.

The BST ordering property is left subtree less than the node, right subtree greater, with a convention chosen for duplicates - option B states the form that admits them. Option A has the inequality reversed. Option C is the AVL balance condition, which is an extra constraint on a self-balancing BST, not part of the BST definition. Option D describes a full binary tree.

Questions 21 to 30

21. Which sorting algorithm uses a divide-and-conquer strategy by recursively splitting the list, sorting sub-lists, and then combining them?

- A) Insertion Sort
- B) Bubble Sort
- C) Selection Sort
- D) Merge Sort

22. Which data structure is commonly used to implement an efficient Symbol Table due to its fast average lookup time?

- A) Stack
- B) Linked List
- C) Hash Table
- D) Queue

23. Which of the following is a common technique used to resolve hash collisions?

- A) Rehashing (increasing table size)
- B) Separate Chaining (using linked lists at each index)
- C) Deleting the colliding key
- D) Ignoring the collision

24. Which complexity case usually provides the most important information for selecting an algorithm for general use?

- A) Best Case Complexity
- B) Average Case Complexity
- C) Worst Case Complexity
- D) Atomized Complexity

25. Space Complexity of an algorithm measures:

- A) The size of the executable code file.
- B) The total memory used by the algorithm, including input, output, and auxiliary space, as a function of input size.
- C) The amount of disk space required to store the algorithm's output.
- D) The number of variables used in the algorithm's implementation.

26. What is the Big-Oh time complexity of searching for an element in a sorted array of size n using Binary Search?

- A) $O(1)$
- B) $O(\log n)$
- C) $O(n)$

D) $O(n \log n)$

27. If an algorithm has time complexity $T(n) = O(g(n))$, which of the following best describes it?

- A) $T(n) \leq c \cdot g(n)$ for all $n > n_0$, for some constants $c > 0$ and $n_0 \geq 0$
- B) $T(n) \geq c \cdot g(n)$ for all $n > n_0$, for some constants $c > 0$ and $n_0 \geq 0$
- C) $c_1 \cdot g(n) \leq T(n) \leq c_2 \cdot g(n)$ for all $n > n_0$, for some constants $c_1, c_2 > 0$ and $n_0 \geq 0$.
- D) $T(n) = g(n)$ for all large n

28. The Greedy Method for algorithm design makes choices that seem best at:

- A) The end of the algorithm
- B) Random points during execution
- C) The current moment (locally optimal)
- D) The beginning of the algorithm only

29. A key difference between Divide and Conquer and Dynamic Programming is that:

- A) Divide and Conquer uses recursion, while Dynamic Programming uses iteration.
- B) Dynamic Programming often deals with overlapping sub-problems, while Divide and Conquer sub-problems are typically independent.
- C) Divide and Conquer is only used for sorting, while Dynamic Programming is used for optimization.
- D) Dynamic Programming guarantees optimality, while Divide and Conquer does not.

30. Which of the following problems is canonically solved using Dynamic Programming?

- A) Sorting an array using Bubble Sort.
- B) Finding the shortest path in an unweighted graph using BFS.
- C) Calculating the n th Fibonacci number efficiently.
- D) Searching in a Binary Search Tree.

▼ **ANSWERS TO QUESTIONS 21-30 BELOW** ▼

Finish all ten before turning the page.

Answers - Questions 21 to 30

21. Answer: D) Merge Sort

Merge Sort splits at the midpoint, sorts each half recursively, then merges the sorted halves - the combine step named in the question. Quick Sort is also divide and conquer, but it does its work in the partition step and has no combine step, so the phrase "combining them" points specifically to Merge Sort. The other three options are quadratic iterative sorts.

22. Answer: C) Hash Table

A hash table gives $O(1)$ average-case insertion and lookup, which is what a compiler needs since it consults the symbol table for every identifier occurrence. Balanced binary search trees are also used when ordered traversal is required, at $O(\log n)$.

23. Answer: B) Separate Chaining (using linked lists at each index)

Separate chaining stores all keys hashing to the same index in a linked list, or a tree, at that bucket. The other standard family is open addressing - linear probing, quadratic probing and double hashing. Rehashing into a larger table is a response to a high load factor rather than a technique for resolving an individual collision, and the last two options lose data.

24. Answer: C) Worst Case Complexity

Worst-case complexity is normally treated as the most important because it is a guarantee: it bounds the running time for every input, which matters when an algorithm must be depended on, and unlike the average case it needs no assumption about the input distribution. Average-case analysis is arguably more informative in practice, and option D is not a real term.

Ambiguous as printed - see the explanation above for what the official key most likely records. Worth flagging to your tutor.

25. Answer: B) The total memory used by the algorithm, including input, output, and auxiliary space, as a function of input size.

Space complexity is total memory as a function of input size. It is worth distinguishing it from auxiliary space, which counts only the extra memory beyond the input - Merge Sort has $O(n)$ auxiliary space, whereas an in-place sort has $O(1)$.

26. Answer: B) $O(\log n)$

Each comparison discards half the remaining range, so the number of steps is the number of times n can be halved, that is $\log_2 n$. The recurrence $T(n) = T(n/2) + O(1)$ solves to $O(\log n)$. Binary search requires the array to be sorted, which is why an unsorted array needs a linear scan.

27. Answer: A) $T(n) \leq c \cdot g(n)$ for all $n > n_0$, for some constants $c > 0$ and $n_0 \geq 0$

That is the formal definition of Big-Oh, an asymptotic upper bound. Option B defines Omega, the lower bound, and option C defines Theta, the tight bound.

28. Answer: C) The current moment (locally optimal)

A greedy algorithm makes the locally optimal choice at each step and never reconsiders it. Whether that yields a globally optimal answer depends on the problem having the greedy-choice property and optimal substructure.

29. Answer: B) Dynamic Programming often deals with overlapping sub-problems, while Divide and Conquer sub-problems are typically independent.

That is the defining distinction. Because DP sub-problems overlap, DP stores their solutions - by memoisation or by a table - so each is solved once; naive divide and conquer on overlapping sub-problems recomputes them, which is why recursive Fibonacci is exponential. Option A is wrong because DP can be recursive with memoisation and divide and conquer can be written iteratively.

30. Answer: C) Calculating the nth Fibonacci number efficiently.

Fibonacci is the standard teaching example: $F(n) = F(n-1) + F(n-2)$ has heavily overlapping sub-problems, so memoisation or a bottom-up table reduces exponential time to $O(n)$. Other canonical DP problems are 0/1 knapsack, longest common subsequence, matrix chain multiplication and Floyd-Warshall.

Questions 31 to 40

31. What is the typical time complexity for both BFS and DFS traversal on a graph represented using an adjacency list, where V is the number of vertices and E is the number of edges?

- A) $O(V \log E)$
- B) $O(V * E)$
- C) $O(V^2)$
- D) $O(V + E)$

32. Which tree traversal visits the nodes in the following order? Left Subtree $\rightarrow$ Right Subtree $\rightarrow$ Root

- A) Inorder Traversal
- B) Preorder Traversal
- C) Postorder Traversal
- D) Level Order Traversal

33. What does it mean for a decision problem to be in the complexity class NP (Non-deterministic Polynomial time)?

- A) The problem can be solved in polynomial time by a standard deterministic algorithm.
- B) The problem cannot be solved in polynomial time.
- C) A proposed solution (certificate) for the problem can be verified for correctness in polynomial time.
- D) The problem requires exponential time to solve.

34. What is the significance of proving that $P = NP$?

- A) It would mean all problems currently considered intractable (like most NPC problems) could be solved efficiently (in polynomial time).
- B) It would prove that quantum computers are not possible.
- C) It would imply that verification is fundamentally harder than solving.
- D) It would mean that greedy algorithms are always optimal.

35. What is the 2's complement of the 8-bit binary number 1101 0010?

- A) 0010 1101
- B) 0010 1110
- C) 1101 0011
- D) 0010 1111

36. The Boolean expression $A + (AB)'$ is equivalent to:

- A) 0
- B) 1
- C) $(AB)'$
- D) $(A+B)'$

37. Which type of flip-flop ensures that the output changes only once per clock edge, completely avoiding race conditions?

- A) SR Flip-Flop
- B) JK Flip-Flop
- C) D Flip-Flop
- D) T Flip-Flop

38. A sequential circuit's output depends on:

- A) Only current inputs
- B) Current inputs and past states
- C) Only past states
- D) External clocks

39. How many states does a ring counter with 5 flip-flops has?

- A) 0
- B) 5
- C) 27
- D) 32

40. Which circuit converts 2^n inputs to n outputs?

- A) Decoder
- B) Encoder
- C) Multiplexer
- D) Demultiplexer

▼ **ANSWERS TO QUESTIONS 31-40 BELOW** ▼

Finish all ten before turning the page.

Answers - Questions 31 to 40

31. Answer: D) $O(V + E)$

Each vertex is visited once and each edge is examined once - twice in an undirected graph - so the total is $O(V + E)$. With an adjacency matrix instead, scanning every row for neighbours makes both traversals $O(V^2)$.

32. Answer: C) Postorder Traversal

The name states where the root is visited: preorder is root, left, right; inorder is left, root, right; postorder is left, right, root. Postorder is used for deleting a tree and for evaluating expression trees, since children must be handled before their parent.

33. Answer: C) A proposed solution (certificate) for the problem can be verified for correctness in polynomial time.

NP is the class of decision problems whose yes-instances have a certificate verifiable in polynomial time. Note what NP does not mean: it does not mean not polynomial. P is a subset of NP, so every easily solved problem is also in NP.

34. Answer: A) It would mean all problems currently considered intractable (like most NPC problems) could be solved efficiently (in polynomial time).

If $P = NP$ then every problem whose solution can be verified quickly can also be solved quickly, so every NP-complete problem - travelling salesman, satisfiability, graph colouring - would have a polynomial algorithm. It would also break most public-key cryptography, which relies on certain problems being hard to solve but easy to verify.

35. Answer: B) 0010 1110

Invert every bit to get the one's complement, 0010 1101, then add 1, giving 0010 1110. Option A is the one's complement, which is the common trap - the final increment has been forgotten.

36. Answer: B) 1

By De Morgan, $(AB)' = A' + B'$, so the expression is $A + A' + B'$. Since $A + A' = 1$ and $1 + \text{anything} = 1$, the whole expression is 1 regardless of the inputs - a tautology.

37. Answer: C) D Flip-Flop

An edge-triggered D flip-flop simply transfers D to Q at the clock edge. It has no input combination that causes toggling, so the race-around condition cannot arise, and it has no forbidden state as the SR does. Race-around is specifically the JK defect, occurring when $J = K = 1$ in a level-triggered device; a master-slave JK also limits the output to one change per cycle, so B is defensible, but D is the type that avoids the problem by construction. T is a JK with the inputs tied together and inherits the same concern.

Ambiguous as printed - see the explanation above for what the official key most likely records. Worth flagging to your tutor.

38. Answer: B) Current inputs and past states

That is the definition of a sequential circuit: it contains memory elements, so the output is a function of the present input and the stored state, which encodes the history of inputs. A circuit whose output depends on current inputs alone is combinational.

39. Answer: B) 5

A ring counter circulates a single 1 through the flip-flops, so an n -bit ring counter has exactly n valid states - here 5. Option D, 32, is 2^5 , which is the count for a binary counter using the same five flip-flops; a ring counter trades that coding efficiency for simple decoding, since each state is identified by one bit. A twisted-ring or Johnson counter would give $2n = 10$ states.

40. Answer: B) Encoder

An encoder compresses 2^n input lines into an n -bit code, so a single active input produces its binary index. A decoder does the reverse, n inputs to 2^n outputs. A multiplexer selects one of many inputs onto a single output line, and a demultiplexer routes one input to one of many outputs.

Questions 41 to 50

41. What is the propagation delay of a 4-bit ripple carry adder with 2ns per full adder?

- A) 2ns
- B) 4ns
- C) 8ns
- D) 16ns

42. What is the output of a 4-bit binary-to-Gray code converter for input 1100?

- A) 1100
- B) 1010
- C) 0110
- D) 1001

43. Microprogrammed control units use:

- A) Control words stored in memory
- B) Hardwired logic gates
- C) Quantum Computing
- D) Analog Circuits

44. Which addressing mode calculates the operand address as: (Register) + Offset

- A) Direct
- B) Base-Indexed
- C) Relative
- D) Implied

45. Pipelining improves performance by:

- A) Increasing clock speed
- B) Overlapping instruction execution
- C) Adding more ALUs
- D) Reducing memory size

46. Which mapping technique divides cache into sets?

- A) Direct Mapped
- B) Set-Associative
- C) Fully Associative
- D) Sector Mapped

47. A Memory-mapped I/O:

- A) treats I/O devices as memory locations
- B) uses separate I/O instructions

- C) is slower than DMA
- D) requires no address bus

48. Which loader function resolves external references between object files?

- A) Absolute Loader
- B) Linking Loader
- C) Bootstrap Loader
- D) Relocating Loader

49. Which parsing technique uses a stack and parsing table?

- A) Recursive descent
- B) LR parsing
- C) LL parsing
- D) Operator precedence

50. What is the primary goal of "offline-first" design?

- A) Ensure functionality without internet.
- B) Reduce app size.
- C) Increase animation speed.
- D) Minimize permissions.

▼ ANSWERS TO QUESTIONS 41-50 BELOW ▼

Finish all ten before turning the page.

Answers - Questions 41 to 50

41. Answer: C) 8ns

In a ripple carry adder the carry must propagate through each stage in turn, so the total delay is the number of stages times the per-stage delay: $4 \times 2 = 8$ ns. This linear growth is what carry look-ahead adders exist to avoid.

42. Answer: B) 1010

The rule is that the most significant Gray bit equals the most significant binary bit, and each remaining Gray bit is the XOR of adjacent binary bits. With $B = 1100$: $G_3 = 1$; $G_2 = 1 \text{ xor } 1 = 0$; $G_1 = 1 \text{ xor } 0 = 1$; $G_0 = 0 \text{ xor } 0 = 0$. So the Gray code is 1010.

43. Answer: A) Control words stored in memory

A microprogrammed control unit stores a microprogram of control words, or microinstructions, in a control memory or control store, and each machine instruction is executed by fetching a sequence of them. It is slower than hardwired control but far easier to design and modify, which is why it suits complex CISC instruction sets.

44. Answer: B) Base-Indexed

A register contributing a base address plus a constant offset is base, or base-indexed, addressing - used for array elements, record fields and stack frames. Relative addressing is the special case where the register is the program counter, which is why it is not the general answer here.

45. Answer: B) Overlapping instruction execution

Pipelining divides instruction execution into stages - fetch, decode, execute, memory, write-back - so several instructions occupy different stages simultaneously. Latency per instruction is not reduced, and may rise slightly, but throughput increases, ideally approaching one instruction per cycle.

46. Answer: B) Set-Associative

Set-associative mapping divides the cache into sets of k lines each; a memory block maps to one set and may occupy any line within it. It is the compromise between direct mapping, where each block has exactly one possible line, and fully associative mapping, where a block may go anywhere.

47. Answer: A) treats I/O devices as memory locations

Memory-mapped I/O assigns device registers addresses within the ordinary memory address space, so the same load and store instructions access them. The alternative, isolated or port-mapped I/O, uses a separate address space and dedicated IN and OUT instructions, which is option B.

48. Answer: B) Linking Loader

A linking loader combines linking with loading: it resolves external symbol references between object modules and loads the result in one pass, so linking happens each time the program is run. An absolute loader places a module at a fixed address with no relocation, a relocating loader adjusts addresses but does not link, and a bootstrap loader loads the operating system at start-up.

49. Answer: B) LR parsing

LR parsing is the shift-reduce technique that maintains an explicit stack of states and symbols and consults ACTION and GOTO tables - the canonical table-and-stack parser, and the one produced by Yacc and Bison. Note that predictive LL(1) parsing is also table driven with an explicit stack, so option C is defensible on a literal reading; the intended answer is LR. Recursive descent uses the implicit call stack and no table.

Ambiguous as printed - see the explanation above for what the official key most likely records. Worth flagging to your tutor.

50. Answer: A) Ensure functionality without internet.

Offline-first treats the local store as the source of truth so the application remains usable without connectivity, synchronising with the server when a connection returns. It assumes the network is unavailable rather than treating that as an error case.

Questions 51 to 60

51. Which of the following is NOT a common type of operating system?

- A) Batch operating system
- B) Real-time operating system
- C) Network operating system
- D) Compiler operating system

52. The invention of Integrated Circuits (ICs) significantly influenced the development of which generation of operating systems?

- A) First Generation
- B) Second Generation
- C) Third Generation
- D) Fourth Generation

53. What is a primarily defining characteristic of fifth-generation operating systems (which are still evolving)?

- A) Primarily focused on batch processing
- B) Emphasis on centralized mainframe computing
- C) Integration of artificial intelligence (AI) and natural language processing
- D) Limited support for networking and distributed systems

54. Which OS component is responsible for managing the CPU's allocation to different processes?

- A) Memory manager
- B) File system manager
- C) Device manager
- D) Process manager

55. What is a Process Control Block (PCB)?

- A) A hardware component that controls the CPU
- B) A data structure used by the OS to store information about a process
- C) A section of memory used to store program code
- D) A signal sent to a process to terminate it

56. What is the purpose of the time quantum in Round Robin scheduling?

- A) To determine the priority of a process
- B) To limit the execution time of a process in each turn
- C) To indicate the remaining execution time of a process
- D) To decide which process should be executed next based on its arrival time

57. Which of the following is a common technique for handling I/O operations?

- A) Paging

- B) Segmentation
- C) Polling
- D) Swapping

58. Which of the following is a common security mechanism implemented in operating systems for controlling access to resources?

- A) CPU scheduling algorithms
- B) Memory paging
- C) Access control lists (ACLs)
- D) File system mounting

59. Which of the following is NOT a fundamental characteristic of a system?

- A) Components
- B) Boundaries
- C) Isolation
- D) Environment

60. Which SDLC model emphasizes iterative development and early customer feedback?

- A) Waterfall
- B) Spiral
- C) Big Bang
- D) V-Model

▼ **ANSWERS TO QUESTIONS 51-60 BELOW** ▼

Finish all ten before turning the page.

Answers - Questions 51 to 60

51. Answer: D) Compiler operating system

There is no such category as a compiler operating system - a compiler is an application, not a class of OS. Batch, real-time and network operating systems are all standard categories, along with time-sharing, distributed, embedded and mobile.

52. Answer: C) Third Generation

The generations track hardware: first used vacuum tubes with plugboards and no real OS; second used transistors with batch systems; third, roughly 1965 to 1980, used integrated circuits and brought multiprogramming, time-sharing and spooling, as in OS/360 and Multics; fourth used LSI and VLSI microprocessors and brought personal-computer and network operating systems.

53. Answer: C) Integration of artificial intelligence (AI) and natural language processing

The fifth generation is characterised by AI integration, natural language interaction, cloud and distributed operation, and intelligent automation. The other three options describe earlier generations.

54. Answer: D) Process manager

The process manager, through the scheduler and dispatcher, decides which process runs and performs the context switch. The memory manager allocates address space, the file system manager handles files and directories, and the device manager drives peripherals.

55. Answer: B) A data structure used by the OS to store information about a process

The PCB, or task control block, is the operating system's record of a process. It holds the process id, state, program counter, CPU register contents, scheduling information, memory-management information such as page-table pointers, accounting data, and the I/O and open-file list. It is what is saved and restored on a context switch.

56. Answer: B) To limit the execution time of a process in each turn

The time quantum is the maximum interval a process may hold the CPU before being preempted and returned to the tail of the ready queue. Its size is the critical tuning parameter: too large and Round Robin degenerates into FCFS, too small and context-switching overhead dominates.

57. Answer: C) Polling

Polling, or programmed I/O, has the CPU repeatedly test a device status register until the device is ready. The alternatives are interrupt-driven I/O and DMA. The other three options are all memory-management techniques, not I/O techniques.

58. Answer: C) Access control lists (ACLs)

An ACL attaches to an object and lists which subjects may perform which operations on it. The alternative formulation is a capability list, attached to the subject rather than the object. Paging provides isolation as a side effect but is not an access-control mechanism.

59. Answer: C) Isolation

General systems theory characterises a system by its components or elements, its boundary separating it from its surroundings, its environment, its interfaces, its inputs and outputs, and its purpose. Isolation is not among them, and is in fact contrary to the idea, since a system interacts with its environment across its boundary.

60. Answer: B) Spiral

The spiral model, due to Boehm, proceeds through repeated cycles of planning, risk analysis, engineering and customer evaluation, with a prototype produced and reviewed in each cycle. Waterfall and the V-model are strictly sequential, and Big Bang has no defined process at all.

Questions 61 to 70

61. Agile methodologies, like Scrum, primarily focuses on:
- A) A risk-driven approach with multiple iterations of prototyping.
 - B) A linear, sequential development process with formal reviews at each stage.
 - C) Comprehensive documentation and strict adherence to plans.
 - D) Flexible responses to change, collaboration, and delivering working software frequently.
62. Which cost estimation method among the following relies on historical data from similar past projects?
- A) Algorithmic models (such as COCOMO)
 - B) Expert Judgement
 - C) Analogy
 - D) Bottom-up Estimation
63. What is the purpose of creating a Software Requirements Specification (SRS) document?
- A) To provide a high-level overview of the project for marketing purposes.
 - B) To serve as a detailed agreement between the stakeholders and the development team on what the software will do.
 - C) To guide the testing team in designing test cases after the software is developed.
 - D) To document the technical design choices made during the implementation phase.
64. Which of the following is a non-functional requirement?
- A) The system shall allow users to log in with a username and password.
 - B) The system shall generate a report of sales data.
 - C) The system shall respond to user requests within 2 seconds.
 - D) The system shall allow administrators to manage user accounts.
65. Testing the interactions and interfaces between different modules or components of the software is called:
- A) Unit testing
 - B) Integration testing
 - C) System testing
 - D) Regression testing
66. What is the primary focus of Software Quality Assurance (QA)?
- A) Writing efficient and bug-free code.
 - B) Testing the software after it has been developed.
 - C) Establishing and maintaining processes and standards to ensure the quality of the software product.
 - D) Deploying the software to the end-users.

67. Which of the following is the most critical factor to consider when scheduling requirement analysis in a software project?

- A) Availability of stakeholders for interviews and feedback.
- B) Completion of the final UI/UX design mockups.
- C) Selection of the programming language for development.
- D) Finalization of the deployment infrastructure.

68. A view in a database system provides:

- A) A physical storage structure for the data.
- B) A way to define the overall schema of the database.
- C) A level of abstraction that allows different users to have personalized perspectives of the database.
- D) The actual implementation details of the database.

69. Which of the following is a type of conceptual data model?

- A) Relational Model
- B) Network Model
- C) Hierarchical Model
- D) Entity-Relationship (ER) Model

70. Logical data independence refers to the ability to change the:

- A) Physical schema without affecting the application programs.
- B) Conceptual schema without affecting the application programs.
- C) View schema without affecting the physical schema.
- D) Internal schema without affecting the external schema.

▼ ANSWERS TO QUESTIONS 61-70 BELOW ▼

Finish all ten before turning the page.

Answers - Questions 61 to 70

61. Answer: D) Flexible responses to change, collaboration, and delivering working software frequently.

That paraphrases the Agile Manifesto, which values individuals and interactions over processes and tools, working software over comprehensive documentation, customer collaboration over contract negotiation, and responding to change over following a plan. Option A describes the spiral model, B the waterfall, and C is what Agile was a reaction against.

62. Answer: C) Analogy

Estimation by analogy takes the actual cost of one or more completed comparable projects and adjusts for known differences. Algorithmic models also use historical data, but indirectly, to calibrate a formula; the direct comparison of one project with another is what analogy means.

63. Answer: B) To serve as a detailed agreement between the stakeholders and the development team on what the software will do.

The SRS is the agreed statement of what the system must do, and it functions as the contract between customer and developer. It should specify what, not how - design decisions belong in the design document, which rules out option D. It does drive test-case design, but that is a consequence rather than its purpose.

64. Answer: C) The system shall respond to user requests within 2 seconds.

A functional requirement says what the system does; a non-functional requirement says how well it does it. A two-second response time is a performance quality attribute. The other three all describe features. Other non-functional categories are reliability, availability, security, usability, scalability, maintainability and portability.

65. Answer: B) Integration testing

Integration testing verifies that separately developed modules work together across their interfaces. Unit testing checks a module in isolation, system testing checks the complete product against its requirements, and regression testing re-runs existing tests to confirm that a change has broken nothing.

66. Answer: C) Establishing and maintaining processes and standards to ensure the quality of the software product.

QA is process oriented and preventive: it defines standards, procedures, reviews and audits so that defects are avoided. Quality Control, of which testing is part, is product oriented and detective, finding defects that already exist. The distinction between QA and QC is the point of this question.

67. Answer: A) Availability of stakeholders for interviews and feedback.

Requirements come from stakeholders, so their availability is the binding constraint on the schedule for this phase - a requirements activity cannot proceed without the people who hold the knowledge. The other three are downstream activities that depend on the requirements rather than preceding them.

68. Answer: C) A level of abstraction that allows different users to have personalized perspectives of the database.

A view is a virtual table defined by a stored query. It belongs to the external level of the three -schema architecture, giving each user or application a tailored subset or transformation of the data, which also serves security by hiding columns and rows. It stores no data itself, unless it is a materialised view.

69. Answer: D) Entity-Relationship (ER) Model

The ER model is a high-level conceptual model: it describes entities, attributes and relationships independently of how they will be stored, which is why it is used at the design stage and then mapped to tables. The relational, network and hierarchical models are logical, or representational, models tied to an implementation approach.

70. Answer: B) Conceptual schema without affecting the application programs.

Logical data independence is immunity of the external schemas and application programs to changes in the conceptual schema, such as adding a table or a column. Option A describes physical data independence. Logical data independence is the harder of the two to achieve.

Questions 71 to 80

71. Which of the following is NOT a DDL statement?

- A) CREATE INDEX
- B) ALTER VIEW
- C) INSERT INTO
- D) DROP SCHEMA

72. A derived attribute is an attribute whose value:

- A) Is stored directly in the database.
- B) Is calculated or determined from the values of other attributes.
- C) Must be provided by the user during data entry.
- D) Uniquely identifies a record in a table.

73. What is the purpose of a composite key?

- A) To act as a foreign key referencing multiple primary keys.
- B) To uniquely identify records using a single attribute with multiple parts.
- C) To uniquely identify records using a combination of two or more attributes.
- D) To improve the performance of database queries.

74. Which of the following is NOT a typical step in query processing?

- A) Parsing
- B) Optimization
- C) Compilation
- D) Execution

75. Relationships between tables in a relational database are typically established using:

- A) Primary keys only.
- B) Foreign keys only.
- C) A combination of primary and foreign keys.
- D) Implicit links based on data values.

76. Which of the following is a key advantage of a distributed database system?

- A) Increased complexity in data management
- B) Lower reliability compared to centralized systems
- C) Improved scalability and availability
- D) Reduced data security

77. What is a "protocol" in the context of computer networks?

- A) A physical device used to connect computers.
- B) A set of rules and procedures that govern communication between network devices.

- C) A type of network cable used for high-speed data transfer.
- D) A software application used to monitor network performance.

78. In which topology does each device have a direct connection to every other device in the network?

- A) Star topology
- B) Ring topology
- C) Bus topology
- D) Mesh topology

79. A device that operates at the data link layer of the OSI model and uses MAC addresses to forward data within a LAN is a:

- A) Router
- B) Switch
- C) Hub
- D) Modem

80. Which transmission medium typically offers the highest bandwidth and lowest attenuation over long distances?

- A) Coaxial cable
- B) Unshielded Twisted-Pair (UTP) cable
- C) Microwave
- D) Fiber-optic cable

▼ **ANSWERS TO QUESTIONS 71-80 BELOW** ▼

Finish all ten before turning the page.

Answers - Questions 71 to 80

71. Answer: C) INSERT INTO

INSERT is Data Manipulation Language: it changes the data, not the structure. DDL - CREATE, ALTER, DROP, TRUNCATE, RENAME - defines and modifies schema objects. The other DML statements are SELECT, UPDATE and DELETE.

72. Answer: B) Is calculated or determined from the values of other attributes.

A derived attribute is computed rather than stored - age from date of birth, or a line total from quantity times unit price. In an ER diagram it is drawn as a dashed ellipse. Storing it as well is a deliberate denormalisation, traded for query speed at the cost of having to keep it consistent.

73. Answer: C) To uniquely identify records using a combination of two or more attributes.

A composite key is a primary key made of two or more columns, used when no single column is unique on its own. The typical case is a junction table resolving a many-to-many relationship, where the key is the pair of foreign keys - for example (student_id, course_id) in an enrolment table.

74. Answer: C) Compilation

The standard steps are parsing and translation, optimisation, code generation, and evaluation or execution. Compilation is not normally listed as a distinct step of query processing - it belongs to programming-language translation. Of the four options it is the odd one out.

75. Answer: C) A combination of primary and foreign keys.

A relationship requires both ends: a primary key in the parent table and a foreign key in the child table whose values reference it. Neither alone is sufficient - a foreign key must have a primary key to point at.

76. Answer: C) Improved scalability and availability

Distributing data across sites allows horizontal scaling and, through replication, continued operation when one site fails, so both scalability and availability improve. Options A, B and D are stated as drawbacks - and increased complexity and a wider attack surface are indeed real costs of distribution, which is why only C is phrased as an advantage.

77. Answer: B) A set of rules and procedures that govern communication between network devices.

A protocol is an agreed set of rules covering message syntax, semantics and timing, so that devices from different manufacturers can interoperate. Examples are TCP, IP, HTTP, FTP and SMTP.

78. Answer: D) Mesh topology

A full mesh connects every node to every other, requiring $n(n-1)/2$ links for n nodes. It is highly fault tolerant because multiple paths exist, but cabling cost grows quadratically, so it is used for backbones rather than for access networks.

79. Answer: B) Switch

A switch is a layer 2 device: it learns MAC addresses, builds a forwarding table, and sends each frame only to the port where the destination lives. A router works at layer 3 with IP addresses, a hub at layer 1 and simply repeats to all ports, and a modem modulates and demodulates signals.

80. Answer: D) Fiber-optic cable

Optical fibre carries light through glass, giving enormous bandwidth, very low attenuation so repeaters can be far apart, complete immunity to electromagnetic interference, and high security since it is difficult to tap without detection. It is the medium of submarine and long-haul backbone links.

Questions 81 to 90

- 81.** Which layer of the OSI model is responsible for logical addressing (IP addresses) and routing of packets?
- A) Data Link Layer
 - B) Network Layer
 - C) Transport Layer
 - D) Session Layer
- 82.** Which protocol in the TCP/IP suite is responsible for providing fast, connectionless data transmission?
- A) IP
 - B) TCP
 - C) HTTP
 - D) UDP
- 83.** What is a Uniform Resource Locator (URL)?
- A) The physical address of a server on the Internet.
 - B) A unique address used to identify a resource on the World Wide Web.
 - C) The protocol used to transfer files over the Internet.
 - D) The language used to create web pages.
- 84.** What is a hop in the context of internet routing?
- A) The passage of a data packet through one router.
 - B) The physical distance between two routers.
 - C) The time it takes for a packet to travel between two routers.
 - D) The number of devices connected to a single router.
- 85.** The deployment of a website to a live server occurs after which phase?
- A) Planning
 - B) Implementation
 - C) Testing
 - D) Maintenance
- 86.** Which HTML tag contains metadata about the HTML document, such as the character set and viewport settings?
- A) <body>
 - B) <title>
 - C) <head>
 - D) <html>
- 87.** Which HTML tag is used to create a hyperlink to another web page or resource?
- A) <link>

- B) <url>
- C) <a>
- D) <href>

88. When a browser requests a PHP file from a server, what is typically the server's first action regarding the PHP code?

- A) It sends the raw PHP code to the browser to be executed.
- B) It executes the PHP code and sends the resulting HTML to the browser.
- C) It ignores the PHP code and only sends the HTML.
- D) It prompts the user to download the PHP file.

89. Which of the following is a valid way to declare a variable in PHP?

- A) variable name = "Hello";
- B) \$variable name = "Hello";
- C) \$variable_name = "Hello";
- D) var variable_name = "Hello";

90. Which of the following is a common way to represent knowledge in a knowledge base?

- A) Rules
- B) Neural Networks
- C) Decision Trees
- D) Genetic Algorithms

▼ ANSWERS TO QUESTIONS 81-90 BELOW ▼

Finish all ten before turning the page.

Answers - Questions 81 to 90

81. Answer: B) Network Layer

Layer 3, the network layer, handles logical addressing with IP, routing and path determination between networks, and fragmentation. Physical addressing with MAC belongs to layer 2, and process-to-process delivery with ports to layer 4.

82. Answer: D) UDP

UDP is the connectionless transport protocol: no handshake, no acknowledgement, no retransmission and an 8-byte header, so it has minimal latency and overhead. It is used for DNS, DHCP, streaming media and games, where timeliness matters more than guaranteed delivery.

83. Answer: B) A unique address used to identify a resource on the World Wide Web.

A URL identifies and locates a web resource. Its parts are the scheme, the host, an optional port, the path, an optional query string and an optional fragment. Option A confuses it with an IP address, which is what DNS resolves the host name to.

84. Answer: A) The passage of a data packet through one router.

One hop is one router traversal. Hop count is the metric used by distance-vector protocols such as RIP, and the IP time-to-live field is decremented once per hop so that a packet caught in a routing loop is eventually discarded. Option C describes latency, not a hop.

85. Answer: C) Testing

Deployment follows testing: the site is built during implementation, verified during testing, then released to production, after which maintenance begins. Deploying before testing is what the phase ordering exists to prevent.

86. Answer: C) <head>

The head element holds document metadata - meta charset, meta viewport, title, link elements for stylesheets and icons, and script tags. None of it is rendered as page content, which appears in body. title is itself one of the elements inside head, so it is a part rather than the container.

87. Answer: C) <a>

The anchor element, with its href attribute, creates a clickable hyperlink. The link element goes in the head and relates external resources such as stylesheets to the document but is not clickable; href is an attribute rather than a tag; and there is no url element.

88. Answer: B) It executes the PHP code and sends the resulting HTML to the browser.

PHP is server-side: the interpreter runs the script on the server and only its output, normally HTML, is sent to the browser. The client never receives the source, which is why database credentials can safely live in a PHP file but not in JavaScript.

89. Answer: C) `$variable_name = "Hello";`

A PHP variable must begin with a dollar sign, followed by a letter or underscore, then letters, digits or underscores - and it may not contain a space. Option A lacks the dollar sign, B contains a space, and var is JavaScript syntax, not PHP.

90. Answer: A) Rules

Production rules of the form IF condition THEN conclusion are the classic knowledge-representation formalism for a knowledge base, alongside semantic networks, frames, logic and ontologies. The other three options are learning or search techniques rather than representation schemes, and neural networks in particular store knowledge in weights that cannot be read as explicit knowledge.

Questions 91 to 100

91. Ontologies in knowledge representation primarily focus on:

- A) Defining the steps in an algorithm.
- B) Establishing a formal and explicit specification of a shared conceptualization.
- C) Storing large amounts of factual data.
- D) Translating natural language into machine code.

92. Which of the following is a common algorithm used in inductive learning for classification tasks?

- A) Back propagation
- B) Q-learning
- C) Decision trees
- D) Genetic algorithms

93. What is a primary advantage of Explanation Based Learning (EBL) compared to purely inductive learning methods?

- A) It typically requires significantly less training data.
- B) It is more robust to noisy data.
- C) It can learn effectively even without any prior knowledge.
- D) It always produces more accurate models.

94. Which of the following might be represented using a structured description in image recognition?

- A) The average color intensity of an image
- B) The number of edges detected in an image.
- C) The arrangement of shapes and their spatial relationships within an image.
- D) The overall texture of an image.

95. In speech recognition, the sequence of phonemes (basic units of sound) recognized in a spoken word can be considered a form of:

- A) Numerical description
- B) Symbolic description
- C) Statistical feature vector
- D) Raw audio waveform

96. Hidden Markov Models (HMMs) were historically a significant technique used in:

- A) Image segmentation
- B) Natural language generation
- C) Speech recognition
- D) Anomaly detection

97. Which of the following is a common application area for PROLOG?

- A) Expert systems and knowledge representation

- B) Statistical data analysis
- C) Game development
- D) Web development

98. Which of the following best represents knowledge-level learning in the context of Artificial Intelligence?

- A) Encoding expert medical knowledge into a rule-based system for disease diagnosis, allowing it to reason about symptoms and conditions logically.
- B) Training a neural network through trial-and-error reinforcement learning to optimize robot navigation in a maze.
- C) Memorizing a dataset of historical weather patterns to predict future temperatures without understanding meteorology.
- D) Fine-tuning a deep learning model on labeled images to classify objects without explicit feature engineering.

99. A key challenge with distance-based methods like k-NN in high-dimensional spaces is:

- A) The increased computational cost for storing and searching the data.
- B) The "curse of dimensionality," where distances between points become less meaningful, and data becomes sparse.
- C) The difficulty in choosing an appropriate distance metric.
- D) All of the above.

100. The k-Nearest Neighbours (k-NN) algorithm is a type of:

- A) Linear regression model
- B) Instance-based or memory-based learning algorithm
- C) Tree-based ensemble method
- D) Generative probabilistic model

▼ **ANSWERS TO QUESTIONS 91-100 BELOW** ▼

Finish all ten before turning the page.

Answers - Questions 91 to 100

91. Answer: B) Establishing a formal and explicit specification of a shared conceptualization.

That is Gruber's standard definition. An ontology fixes the concepts of a domain, their properties and the relationships between them, in a form that both machines and people can agree on - which is what makes knowledge shareable and reusable across systems. RDF and OWL are the Semantic Web languages for expressing them.

92. Answer: C) Decision trees

Decision-tree induction, as in ID3, C4.5 and CART, is the classic inductive classification method: it generalises a set of labelled examples into an explicit tree of tests. Back propagation is a training procedure for neural networks rather than a learning paradigm in itself, Q-learning is reinforcement learning, and genetic algorithms are a search and optimisation technique.

93. Answer: A) It typically requires significantly less training data.

EBL can generalise from a single solved example, because it uses a domain theory to explain why that example worked and then generalises the explanation. Inductive methods need many examples to distinguish signal from coincidence. The trade-off is precisely option C reversed: EBL depends on having a domain theory, so it cannot operate without prior knowledge.

94. Answer: C) The arrangement of shapes and their spatial relationships within an image.

A structured, or symbolic, description represents an image as parts together with the relations between them - above, adjacent to, contained in - rather than as a vector of numbers. The other three options are all scalar or statistical features, which are numerical descriptions.

95. Answer: B) Symbolic description

A phoneme is a discrete symbol drawn from a finite alphabet, so a sequence of them is a symbolic representation - one abstraction level above the numerical feature vectors, such as MFCCs, from which it was derived, and two above the raw waveform.

96. Answer: C) Speech recognition

HMMs dominated speech recognition from the 1970s until deep neural networks displaced them. They fit the problem because speech is a sequence whose underlying phonetic states are hidden and must be inferred from acoustic observations, which is exactly what the Viterbi and forward-backward algorithms do.

97. Answer: A) Expert systems and knowledge representation

PROLOG is a declarative logic-programming language: a program is a set of facts and rules, and computation is resolution with unification and backtracking. That makes it a natural fit for expert systems, knowledge representation, theorem proving and natural-language parsing.

98. Answer: A) Encoding expert medical knowledge into a rule-based system for disease diagnosis, allowing it to reason about symptoms and conditions logically.

Knowledge-level learning extends what the system can deduce - it acquires explicit, reasoned-about knowledge that enlarges its deductive closure. Encoding medical rules that support logical reasoning about symptoms does exactly that. The other options are symbol-level or purely statistical: they improve performance on a task without adding explicit knowledge the system can reason with.

99. Answer: D) All of the above.

All three are genuine problems in high dimensions, and they compound each other. The curse of dimensionality is the deepest of them - as dimensions grow, the ratio of nearest to farthest distance approaches 1, so the very idea of a nearest neighbour loses its force - but cost and metric choice are real too, so the inclusive option is correct.

100. Answer: B) Instance-based or memory-based learning algorithm

k-NN stores the training instances and defers all computation to prediction time, comparing a new case against the retained examples. It is therefore instance-based, also called memory-based or lazy learning. It builds no model and estimates no parameters, which is why it is non-parametric.

